

caad



CHENNAI
ACADEMY OF
ARCHITECTURE AND
DESIGN

POONAMALLEE CITY CAMPUS,
CHENNAI.

NATA NEW FORMAT

GUIDED LEARNING RESOURCES

(PREPARED BY EXPERT FACULTY IN ARCHITECTURE EDUCATION)



B.Arch.,

ANNA UNIVERSITY
COUNCELLING CODE

1152

CONTACT

WWW.CAAD.AC.IN

9710554545 | 9710930025

NUMERICAL ABILITY

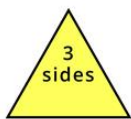
Basic Mathematics and its association with creative thinking.
To unfold a space with use of geometry.

NUMERICAL REASONING

GEOMETRY

REGULAR POLYGON

- Sum of all interior angles of a regular polygon of side n is given by $(2n - 4) 90^\circ$.
- Angle of a regular polygon = $((2n-4)90^\circ)/n$
- Sum of an interior angle and its adjacent exterior angle is 180°
- Sum of all exterior angles of a polygon taken in order is 360° .



Equilateral Triangle



Square



Regular Pentagon



Regular Hexagon



Regular Heptagon



Regular Octagon

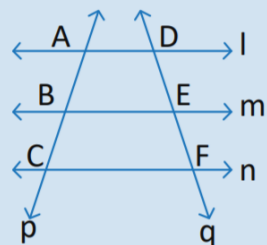


Regular Nonagon



Regular Decagon

PROPERTY OF INTERCEPTS MADE BY THREE PARALLEL LINES

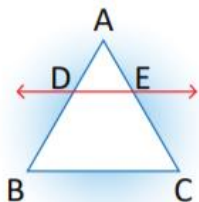


If $AD \parallel BE \parallel CF$,

$$\text{then } \frac{AB}{BC} = \frac{DE}{EF}$$

The ratio of intercepts made on transversal by 3 parallel lines is equal to ratio of corresponding intercepts made on any other transversal of the same parallel lines

BASIC PROPORTIONALITY THEOREM



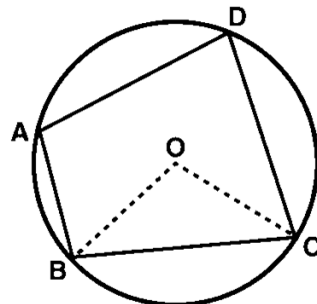
If $DE \parallel BC$, then by BPT

$$\frac{AD}{DB} = \frac{AE}{EC}$$

If a line is parallel; to a side of a triangle which intersects other two sides in distinct points, then the line divides other two sides in proportion.

CYCLIC QUADRILATERAL

In a cyclic quadrilateral, the sum of a pair of opposite angles is 180° (supplementary).



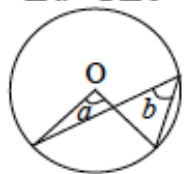
The area of a cyclic quadrilateral is

$$\text{Area} = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

where a, b, c , and d are the four sides of the quadrilateral.

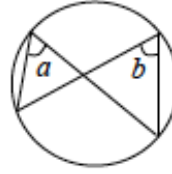
∠ at Centre

$$\angle a = 2 \angle b$$

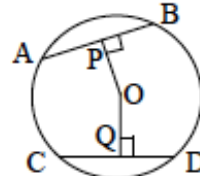


∠s in Same Segment

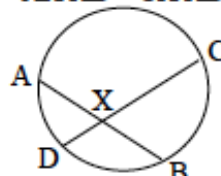
$$\angle a = \angle b$$



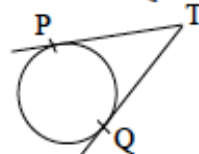
Equal chords equidistant from centre
 $AB = CD \leftrightarrow OP = OQ$



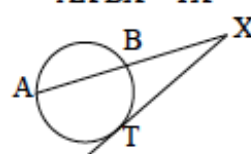
Intersecting Chords Theorem
 $AX \cdot XB = CX \cdot XD$



Tangents from external point
 $TP = TQ$

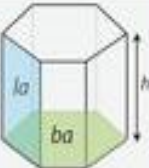
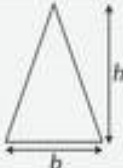
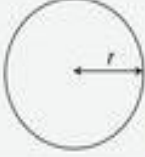
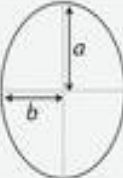
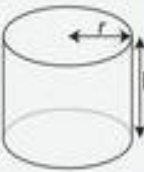


Tangent-Secant Theorem
 $AX \cdot BX = TX^2$



NUMERICAL REASONING

MENSURATION

Two-dimensional plane shapes	Area <i>The measure of how many squares will fit into a shape.</i> Units²	Three-dimensional solid shapes	Surface Area <i>The measure of the area of all outward facing sides.</i> Units²	Volume <i>The measure of how many cubes will fit into a shape.</i> Units³
Square 	Area = a^2 or $a \times a$ Example: $a = 5\text{cm}$ Area = $5^2 = 25\text{cm}^2$	Cube 	Surface Area = $6 \times a^2$ Example: $a = 5\text{cm}$ Surface Area = 150cm^2	Volume = a^3 or $a \times a \times a$ Example: $a = 5\text{cm}$ Volume = 125cm^3
Rectangle 	Area = $w \times h$ Example: $w = \text{width} = 10\text{cm}$ $h = \text{height} = 20\text{cm}$ Area = $10 \times 20 = 200\text{cm}^2$	Prism 	Surface Area = $2 \times ba + la$ Example: $ba = \text{base area} = 20\text{cm}^2$ $la = \text{lateral area (all sides)} = 60\text{cm}^2$ Surface area = $2 \times 20 + 60 = 100\text{cm}^2$	Volume = $ba \times h$ Example: $ba = \text{base area} = 20\text{cm}^2$ $h = \text{height} = 5\text{cm}$ Volume = $20 \times 5 = 100\text{cm}^3$
Triangle 	Area = $b \times h \times 0.5$ Example: $b = \text{base} = 20\text{cm}$ $h = \text{vertical height} = 15\text{cm}$ Area = $20 \times 15 \times 0.5 = 150\text{cm}^2$	Pyramid 	Surface Area = $ba + la$ Example: $ba = \text{base area} = 16\text{cm}^2$ $la = \text{lateral area (all sides)} = 60\text{cm}^2$ Surface area = $16 + 60 = 76\text{cm}^2$	Volume = $ba \times h \times 1/3$ Example: $ba = \text{base area} = 16\text{cm}^2$ $h = \text{height} = 9\text{cm}$ Volume = $16 \times 9 \times 1/3 = 48\text{cm}^3$
Reg Polygon 	Area = $n \times s \times a \times 0.5$ Example: $n = \text{number of sides} = 6$ $\text{length of side} = 5\text{cm}$ $a = \text{apothem} = 15\text{cm}$ Area = $6 \times 5 \times 15 \times 0.5 = 225\text{cm}^2$	R. Polyhedron 	Surface Area = $fa \times s$ Example: $fa = \text{area of one side} = 200\text{cm}^2$ $s = \text{number of sides} = 12$ Surface area = $200 \times 12 = 2400\text{cm}^2$	Example: There is no simple generic formula for working out the volume of a regular polyhedron.
Circle 	Area = $\pi \times r^2$ Example: $\pi = \text{pi} = 3.14$ $r = \text{radius} = 5\text{cm}$ Area = $3.14 \times 5^2 = 3.14 \times 5 \times 5 = 78.5\text{cm}^2$	Sphere 	Surface Area = $4 \times \pi \times r^2$ Example: $r = \text{radius} = 4.5\text{cm}$ Surface area = $4 \times 3.14 \times 20.25 = 254.5\text{cm}^2$ (Approx)	Volume = $4/3 \times \pi \times r^3$ Example: $r = \text{radius} = 4.5\text{cm}$ Volume = $4/3 \times 3.14 \times 4.5^3 = 381.5\text{cm}^3$ (Approx)
Ellipse 	Area = $\pi \times a \times b$ Example: $\pi = \text{pi} = 3.14$ $a = \text{radius of long axis} = 6$ $b = \text{radius short axis} = 4$ Area = $3.14 \times 6 \times 4 = 75.36\text{cm}^2$	Cylinder 	Surface Area = $2\pi rh + 2\pi r^2$ Example: $r = \text{radius} = 5\text{cm}$ $h = \text{height} = 10\text{cm}$ Surface area = $2 \times 3.14 \times 5 \times 10 + 2 \times 3.14 \times 25 = 471\text{cm}^2$	Volume = $\pi \times r^2 \times h$ Example: $r = \text{radius} = 5\text{cm}$ $h = \text{height} = 10\text{cm}$ Volume = $3.14 \times 25 \times 10 = 785\text{cm}^3$ (Approx)

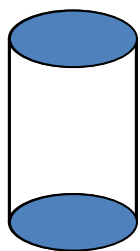
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NUMERICAL REASONING

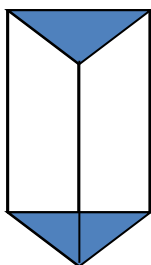
SOLIDS

- Solids having top and base of same shape is a prism

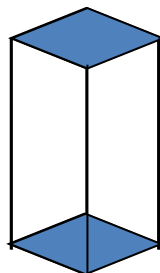
PRISMS



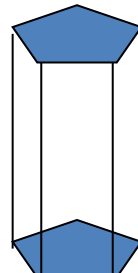
Cylinder



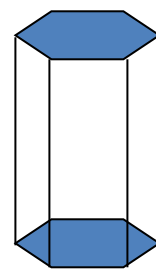
Triangular



Square



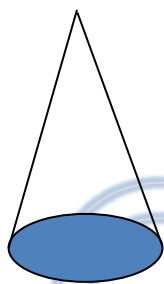
Pentagonal



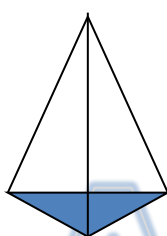
Hexagonal

- Solids having base of some shape and just a point as a top, called apex is a pyramid

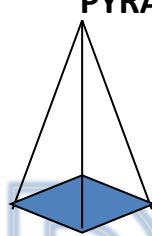
PYRAMIDS



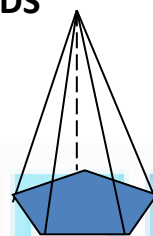
Cone



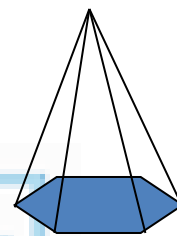
Triangular



Square



Pentagonal



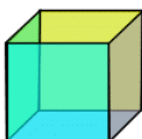
Hexagonal

PLATONIC SOLIDS

TETRAHEDRON
4 triangle faces



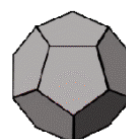
CUBE
6 square faces



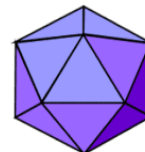
OCTAHEDRON
8 triangle faces



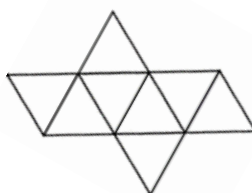
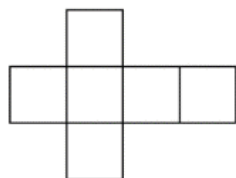
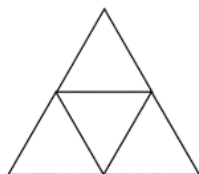
DODECAHEDRON
12 pentagon faces



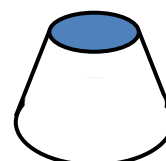
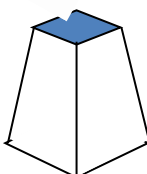
ICOSAHEDRON
20 triangle faces



NETS OF PLATONIC SOLIDS



FRUSTUM OF CONE & PYRAMIDS.
(top & base parallel to each other)

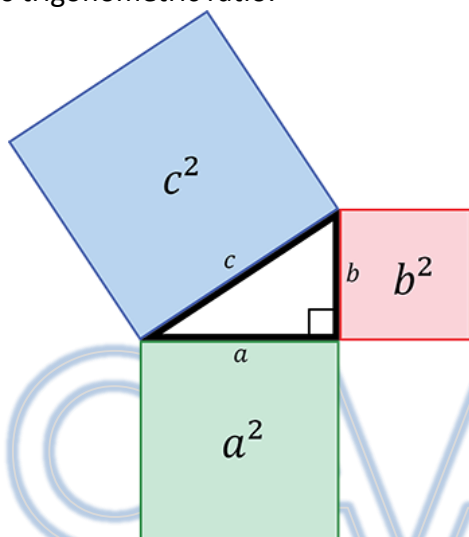
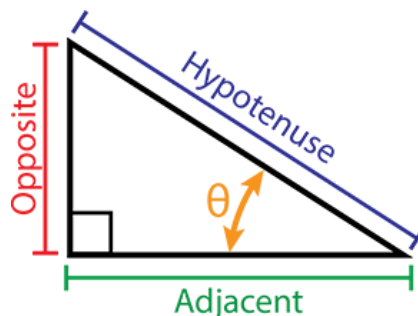


NUMERICAL REASONING

BASICS OF TRIGONOMETRY

TRIGONOMETRIC RATIOS

The most important task of trigonometry is to find the remaining side and angle of a triangle when some of its side and angles are given. This problem is solved by using some ratio of sides of a triangle with respect to its acute angle. These ratio of acute angle are called trigonometric ratio of angle. Let us now define various trigonometric ratio.



$\sin \theta$	= Perpendicular / Hypotenuse
$\cos \theta$	= Adjacent / Hypotenuse
$\tan \theta$	= Perpendicular / Adjacent
$\operatorname{cosec} \theta$	= Hypotenuse / Perpendicular
$\sec \theta$	= Hypotenuse / Adjacent
$\cot \theta$	= Adjacent / Perpendicular

PYTHAGORAS THEOREM

The square on the hypotenuse is equal to the sum of the squares on the other two sides.

$$c^2 = a^2 + b^2$$

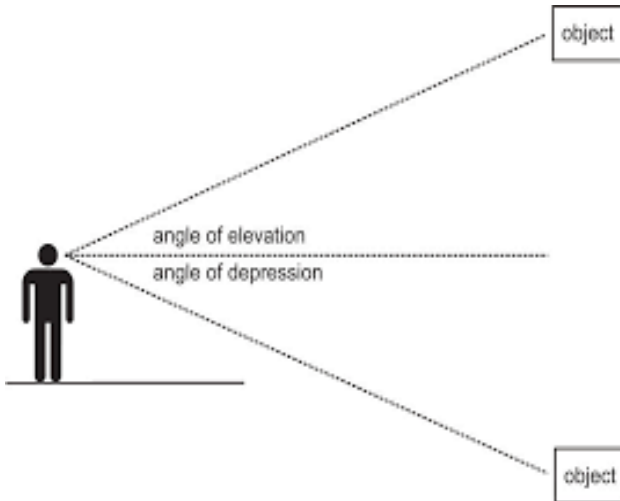
θ T-ratio	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
$\operatorname{cosec} \theta$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
$\cot \theta$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

NUMERICAL REASONING

APPLICATIONS OF TRIGONOMETRY

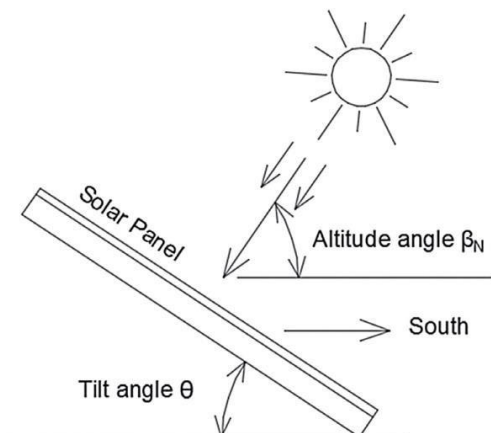
HEIGHT AND DISTANCE

Sometimes, we have to find the height of a tower, building, tree, distance of a ship, width of a river, etc. Though we cannot measure them easily, we can determine these by using trigonometric ratios.



Applications

- Finding heights of towers or buildings or trees
- Finding distance between two non accessible points.
- Measuring fields, lots and areas
- Making walls parallel and perpendicular
- Roof inclination
- To calculate sun shading and light angles.



BEST TILT ANGLE ?

LINE OF SIGHT

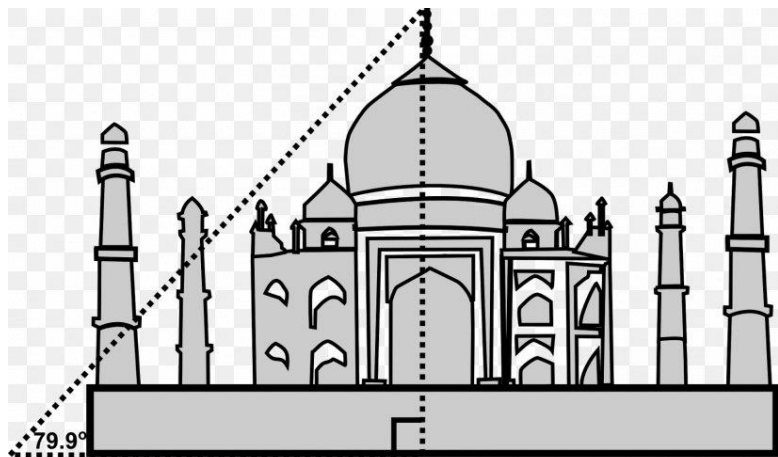
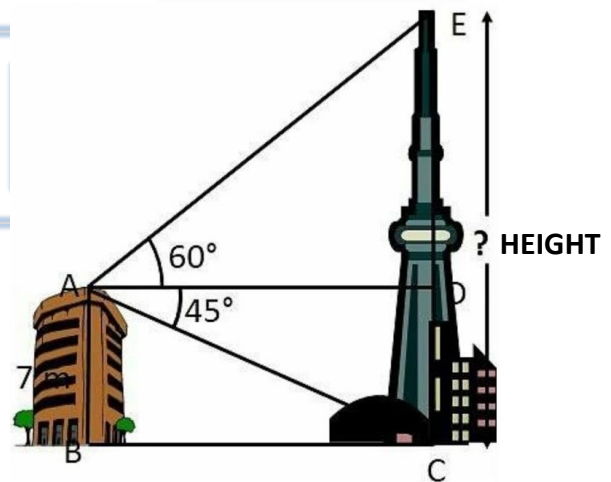
The line of sight or the line of vision is a straight line to the object we are viewing.

ANGLE OF ELEVATION

If the object is above the horizontal from the eye, we have to lift up our head to view the object. In this process, our eye move, through an angle. This angle is called the angle of elevation of the object.

ANGLE OF DEPRESSION

If the object is below the horizontal from the eye, then we have to turn our head downwards to view the object. In this process, our eye move through an angle. This angle is called the angle of depression of the object.



DISTANCE ?

NUMERICAL REASONING

RATIO AND PROPORTION

Ratio and Proportion are explained majorly based on fractions and it is the foundation to understand the various concepts in mathematics as well as in science.

RATIO

When a fraction is represented in the form of $a:b$, then it is a ratio.

PROPORTION

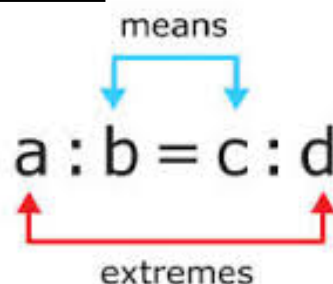
When two ratios are equal, the four quantities composing them are said to be in proportion.

- If , then a, b, c, d are in proportion.
- This is expressed by saying that 'a' is to 'b' as 'c' is to 'd' and the proportion is written as $a:b::c:d$ or $a:b=c:d$
- The terms a and d are called the extremes while the terms b and c are called means.

RATIO

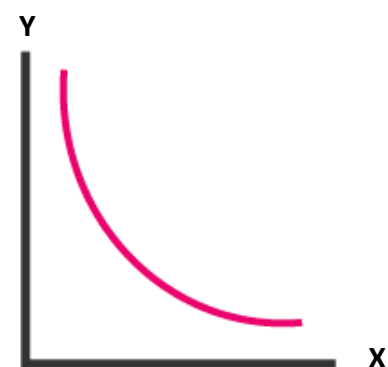
$$a : b = \frac{a}{b} \begin{array}{l} \text{— antecedent} \\ \text{— consequent} \end{array}$$

PROPORTION



DIRECT PROPORTION

$$y = kx, \text{ where } k \text{ is a constant}$$



INVERSE PROPORTION

$$y = k/x, \text{ where } k \text{ is a constant}$$

DIRECT PROPORTION

If on the increase of one quantity, the other quantity increases to the same extent or on the decrease of one, the other decrease to the same extent, then we say that the given two quantities are directly proportional.

Some Examples:

- Work done / Number of men
- Cost / Number of men
- Work / Wages
- Working hour of a machine / Fuel consumed
- Speed / Distance to be covered

INVERSE PROPORTION

If on the increase of one quantity, the other quantity decreases to the same extent or vice versa, then we say that the given two quantities are indirectly proportional.

Some Examples:

- More men / Less time
- Less men / More hours
- More speed / Less time

NUMERICAL REASONING

PERCENTAGE

A percentage is a fraction with denominator hundred, It is denoted by the symbol %.

Numerator of the fraction is called the rate per cent.

POPULATION FORMULA

If the original population of a town is P, and the annual increase is r %, then the population after n years is $P (1 + r/100)^n$ and population before n years = $P/[1 + (r/100)]^n$

If the annual decrease be r%, then the population after n years is $P (1 - r/100)^n$ and population before n years = $P/[1 + (r/100)]^n$

SUCCESSIVE INCREASE OR DECREASE

In the value is increased successively by x% and y% then the final increase is given by

$$[x + y + (xy/100)] \%$$

In the value is decreased successively by x% and y% then the final decrease is given by

$$[-x - y - (xy/100)] \%$$



2-DIMENSIONAL FIGURE AND AREA

If the sides of a triangle, square, rectangle, rhombus or radius of a circle are increased by a%, its area is increased by $[a(a+200)]/100 \%$

If the sides of a triangle, square, rectangle, rhombus or radius of a circle are decreased by a %, its area is decreased by $[a(200-a)]/100 \%$

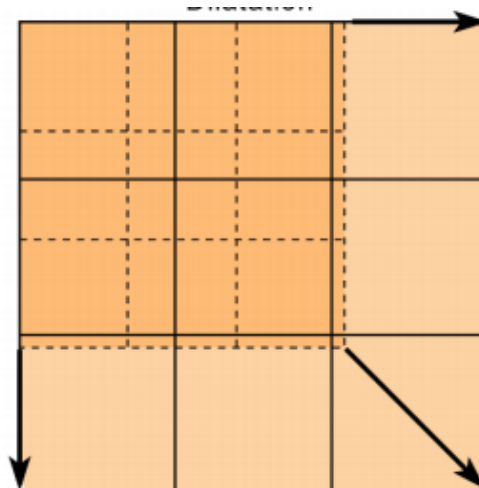


STUDENT AND MARKS

The percentage of passing marks in an examination is x%. If a candidate who scores y marks fails by z marks, then the maximum marks - $M = [100 (y + z)] / x$

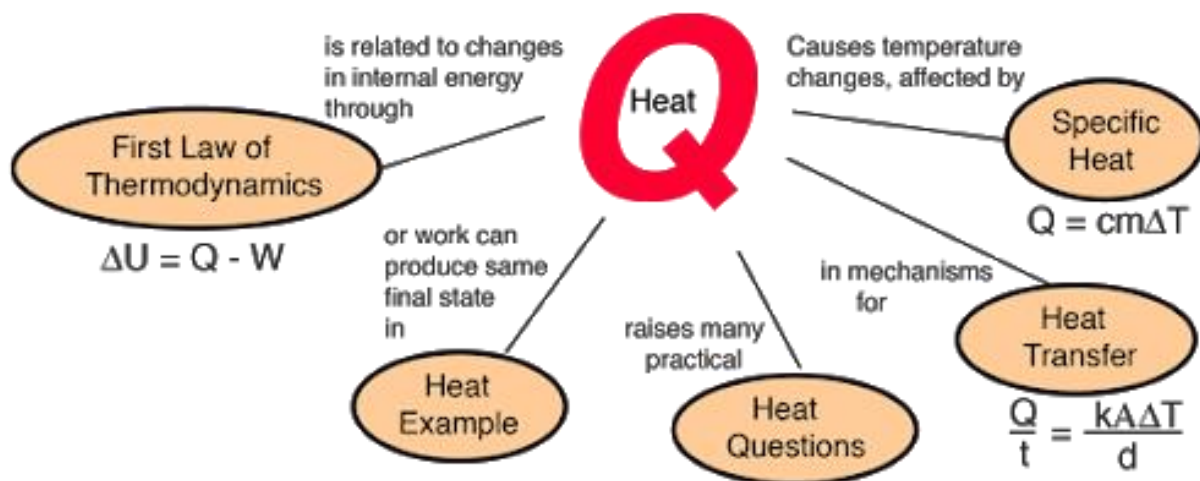
A candidate scoring x% in an examination fails by 'a' marks, while another candidate who scores y% marks gets 'b' marks more than the minimum required passing marks, then the maximum marks - $M = [100 (a+b)] / (y-x)$

In an examination x% and y% students respectively fail in two different subjects while z% students fail in both subjects, then the % age of student who pass in both the subjects will be $\{100-(x + y - z)\} \%$



NUMERICAL REASONING

PHYSICS



CONVERSION OF FARENHEIT TO CELCIUS SCALE

On any temperature scale, $\frac{T - \text{ice point}}{\text{steam point} - \text{ice point}}$ remains constant. So,

$$\left(\frac{T - \text{ice point}}{\text{steam point} - \text{ice point}} \right)_{\text{Fahrenheit scale}} = \left(\frac{T - \text{ice point}}{\text{steam point} - \text{ice point}} \right)_{\text{Celsius scale}} \Rightarrow T_c = \frac{5}{9}(T_f - 32)$$

SPEED, DISTANCE AND TIME TAKEN

Speed is the rate at which any moving body covers a particular distance.

$$\text{SPEED} = \text{DISTANCE} / \text{TIME}$$

RELATIVE SPEED

When two bodies are moving in same direction with speeds s_1 and s_2 respectively, their relative speed is the difference of their speeds.

$$\text{RELATIVE SPEED} = s_1 - s_2$$

When two bodies are moving in opposite direction with speeds s_1 and s_2 respectively, their relative speed is the sum of their speed.

$$\text{RELATIVE SPEED} = s_1 + s_2$$

TRAINS

When two trains with lengths L_1 and L_2 and with speeds S_1 and S_2 respectively, then

- When they are moving in the same direction, time taken by the faster train to cross the slower train = $(L_1 + L_2) / \text{difference of their speeds}$.
- When they are moving in the opposite direction, time taken by the trains to cross each other = $(L_1 + L_2) / \text{sum of their speeds}$.

BOAT AND STREAM

Let the speed of a boat (or man) in still water be X m/sec and the speed of the stream (or current) be Y m/sec. Then,

Speed of boat with the stream (or downstream) = $(X + Y)$ m/sec

Speed of boat against the stream (or upstream) = $(X - Y)$ m/sec

Speed of boat in still water is

$$X = (\text{Downstream} + \text{Upstream}) / 2$$

DIRECTIONS

In our day to day life, we get our direction after seeing the position of sun.

Left turn is a clockwise turn

Right turn is an Anti-clockwise turn

